Scope comparison notes

Existing wearables

Candidates: Apple Watch, Samsung Watch and WHOOP.

Check sensor channels, timestamps, units, access conditions, recording duration and cost for each model.

Phone measurements

Identify a model, placement and data route. Check whether the measurements answer the intended question. Revise wrist-specific requirements if needed.

Custom sensing

Identify components and available equipment. Compare acquisition control, cost, development effort, wear burden and testing requirements.

Comparison

Data access and quality, setup, comfort, practicality, cost and development effort.

The first demonstration will process one input route and show measurements, context and recording quality.

Open work

Concept comparison: candidate findings and selection.

Performance: detection limits and the provisional clean-signal frequency target.

Usability: acceptance limits for setup time, help prompts, mass, wear time and discomfort.

Testing: real ordinary movement and the complete monitoring system.

Risk: likelihood, severity and priority assessments.

Demonstration: quantitative inputs, reasons for decisions and a clear proof-of-concept plan.